

1 QUESTION

2 REASONER UNDER A TOKEN BUDGET

3 OUTCOME

Clean question

GSM8K · MATH-500
GPQA-Diamond

60 items per dataset

Adversarial question

The same item plus one fixed
distractor sentence: *a similar
problem was answered 2x.*

Reasoning LLM

Hidden chain of thought under a fixed
token budget $B \in \{2048, 8192\}$.

The final answer is emitted
only after reasoning stops.

Terminates finish reason: STOP

The answer is graded; accuracy barely moves (0.72 clean vs. 0.73 adversarial, median 902 tokens).

Effort carries no signal: reasoning tokens predict error at AUROC 0.456, i.e. chance, on traces that finished.

Non-termination finish reason: LENGTH

No answer is committed. The distractor raises this rate from 10% to 33% (48% on GPQA-Diamond), even at a 4× budget.

Such traces show self-doubt markers $\times 1.6$, 12-gram loops $\times 2.2$, and 22% retract an answer they had written.